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Novel and familiar object recognition rely on the same ability

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Abstract:

There is recent evidence for a domain-general object recognition ability, called *o*, which is distinct from general intelligence and other cognitive and personality constructs. We extend the study of *o* by characterizing how it generalizes to the ability to recognize familiar objects, and to the ability to make judgments of the average identity of ensembles of objects. We applied latent variable modeling to data collected from a sample of adults (*N*=284), in three different tasks and for six different object domains (three novel and three familiar). The results replicated prior work in finding that on average 88% of the variance of lower-order factors could be accounted by *O* for novel objects. The latent constructs recruited by the higher-order factor for novel objects and for familiar objects were almost perfectly correlated and therefore, functionally identical. A latent factor for ensemble perception shared about 42% of the variance with *o*, suggesting at least strong overlap between abilities supporting judgments about individual objects and ensemble of objects. This work extends the theoretical reach of *o* by showing generalization across two dimensions (familiar vs. novel objects; individual vs. ensemble object perception). With respect to the structure of individual differences in high-level vision, researchers would benefit from accounting for the contribution of *o* when seeking to understand various domain-specific abilities.

Many daily tasks, as part of work, educational activities or hobbies, rely on visual object recognition skills. These skills often depend on domain-specific learning (e.g., having gone through relevant training for fingerprint matching). But a new body of work also points to the relevance of domain-general visual ability in accounting for how easily people can acquire visual object recognition skills and influencing individual differences in performance after training. A common intuition is that we vary less in our visual abilities than in other cognitive skills, but evidence suggests comparable amounts of variation in the general population (Gauthier, 2018). Compared to the abundance of research and applications in the field of general intelligence (e.g., Bouchard & McGue, 1981; Deary, 2001; Spearman, 1927), the systematic study of individual differences in the ability to visually learn, discriminate and recognize objects is a new development. Here, we focus on a domain-general visual object recognition ability thought to reflect individual differences in the creation and manipulation of visual representations for objects (Richler et al., 2019). We build on evidence that a domain-general visual ability, which has been called “O”, can account for a majority of the variance across different tasks and with different categories of novel objects, and is only weakly correlated with general intelligence, g (Richler et al., 2017). We investigate the generalization of this ability along two dimensions that have been proposed to justify the consideration of separate mechanisms within high-level visual processing. One dimension is the processing of familiar vs. novel objects and a second one is the processing of ensembles vs. individual objects. If O is relevant across these dimensions, this would inform these specific debates about distinct mechanisms. But a general O could also provide a unifying construct to guide future research into mechanisms of object recognition (Mollon et al., 2017; Wilmer, 2008), with both clinical (Reavis et al., 2017; Sunday et al., 2017; Trueblood et al, 2017) and occupational applications (Biggs et al., 2013; Davis et al., 2016).

Object Familiarity

We ask whether O, as measured for novel objects, also applies to the recognition of familiar objects. Psychologists have used objects from both familiar and novel categories to test and develop theories of object recognition (Biederman, 1987; Richler & Palmeri, 2014; Tarr et al., 1998; Ullman et al., 2002). Admittedly, novel objects can be compared and related to objects we have seen before, but we define a novel category as one created by experimenters and with which observers have no experience. While we often encounter new instances within familiar categories (e.g., a species of bird we did not know before), our experience with the other instances of a familiar category provides implicit information about the plausible range of feature variability (Ullman, 2007). For example, our experience with cats and dogs prepares us to expect that dogs vary in size and aspect ratio more than cats do, or that there are no cats with very long floppy ears. In contrast, such information is lacking (or at best has to be learned on the fly) for novel categories. Familiarity can refer to a small amount of exposure, to experience with common basic-level categories like birds and houses even when we do not practice individuating them (Rosch et al., 1976) or to a greater degree of experience with individual objects, resulting in behavioral markers such as holistic processing (Richler & Gauthier, 2014).

Familiarity at almost any level can influence object processing, as evidenced by effects on perceptual representations, attentional processing or categorization. Exposure to specific distractors over trials can improve visual search (Mruczek & Sheinberg, 2005) and increase the neural responses to these images (Anderson et al., 2008). Expertise discriminating objects from an entire category, as found in bird watchers or car aficionados, can lead to categorization that is as fast at the individual level (e.g., Canada warbler) as at the basic-level (e.g., bird; Tanaka & Taylor, 1991). A similar effect is found for individually famous objects for which we have a rich semantic network (e.g., Eiffel Tower; Anaki & Bentin, 2009). Object recognition judgments are also influenced by semantic knowledge that we acquire for familiar categories (Gilbert et al., 2006; Lupyan, 2008; Richler et al., 2011), or by the knowledge of how to manipulate some objects (Chua et al., 2017; Herbort & Butz, 2011; Yoon et al., 2002). New semantic information associated with objects influences the relative hemispheric contribution to perceptual judgments (Curby et al., 2004). Performance with familiar objects is often superior in some way to the processing of novel objects, and in some cases may at least suggest qualitative differences in mechanisms. For instance, while working memory capacity is often considered fixed for novel objects, some have proposed that it is not fixed for familiar objects (Brady et al., 2015). In sum, experimental studies provide some evidence that familiarity cannot be ignored in object recognition.

In terms of individual differences, correlations in performance on memory tasks with various familiar categories (e.g., birds, shoes, leaves, motorcycles) tend to be low, around r=.3 (McGugin et al., 2012; Van Gulick et al., 2016), suggesting important influences of domain-specific experience. In contrast, performance on similar tasks with various novel object categories are more strongly correlated, around r=.5 (Richler et al., 2017), suggesting a stronger contribution of a domain-general ability. There is evidence that the higher pairwise correlations for novel than familiar objects may be due to category-specific experience effects that can be averaged out when more categories are used. In a re-analysis of VanGulick et al. (2015), pairwise correlations of performance between 8 tests with familiar objects was about .3, but when each category was correlated with the average of performance for the 7 other categories, the correlation rose to .5. More recently, Richler et al. (2019) used a variety of tasks with novel objects and found evidence for *O.* This higher-order factor accounted for about 89% of the variance in performance with different tasks and object categories. It is not clear whether the stronger evidence for a domain-general ability with novel than with familiar objects is due to a more sophisticated latent variables approach, to confounding effects of experience with familiar categories, or to new mechanisms that come to bear as a function of familiarity.

In the present work, we are concerned with basic-level familiarity, the kind of familiarity that most people have with common objects like chairs, cups or cats. We ask whether familiarity at the basic-level affects the structure of *individual differences* in object recognition. The tasks we use to measure object recognition ability require discriminating between visually similar objects that share a common configuration of parts, so they require subordinate-level discrimination. Familiarity at the basic level could influence individual differences in subordinate-level processing if people use an existing schema when dealing with individual objects from a familiar category. Knowledge about what parts the objects are made of or even about non-visual semantic information may facilitate encoding. As a result, individual differences in the ability to learn such information on the fly could contribute to variability in performance for novel objects more than for familiar objects. Familiar object recognition may also differ from novel object recognition because people vary in their experience individuating objects from familiar object categories. Variability in experience can contribute to variability in performance and thus influence shared variance between categories (for instance inflating correlations in performance for two categories for which experience is correlated, such as birds and leaves). It is unknown to what extent such complicating factors for familiar object recognition actually limits the measurement of the similarity to novel object recognition in a latent variable framework. Facing these challenges, we aim to characterize how the variance that is shared across tasks with familiar categories is related to the variance that is shared across tasks with novel categories.

In exploring the role of object familiarity, we avoid for the present purposes categories like faces and cars, for which most people may have extreme familiarity and a higher degree of practice with subordinate-level discrimination. Recognition abilities for faces and for cars are more independent from general object recognition in people from larger hometowns (i.e., with larger numbers of faces and cars) than in people from smaller hometowns (Sunday et al., 2019). Studying individual differences for such categories may require a different approach than what we use here, although there may well be no qualitative difference between moderate familiarity with objects and such high degrees of experience. One challenge is that in addition to producing lower correlations with most object categories, performance with faces and cars often also dissociate from one another (McGugin et al., 2012; Ćepulić et al., 2018). This makes it challenging because such specialized categories would not load well on a common latent factor.

Individual Objects vs. Object Ensembles

We also ask whether O, as previously measured with tasks tapping into individual object recognition, also contributes to the recognition of ensembles of objects. Up to now, O has been measured with different tasks, including memory tests, in which a few (6) instances of a category are learned and have to be recognized across a series of trials among distractors, and same/different matching tests, in which a much large number of instances are compared in a pairwise manner over a series of trials that do not require long-term memory. Even as they differ in important ways (e.g., reliance on long-term memory), all these tasks require subjects to attend to properties of individual objects. In contrast, so-called “ensemble perception” tasks require attention to groups of objects as subjects make judgments about summary statistics, such as the average or the variance of an ensemble on some dimension (Whitney & Yamanashi Leib, 2018).

Approximately 20 years ago, Ariely (2001) proposed that object ensembles may be represented differently from individual objects. Some experimental studies support this proposal, when ensemble perception dissociates from single item processing in certain ways. For instance, ensemble perception can be above chance when stimulus presentation is so short that individual items cannot be perceived (e.g., Haberman & Whitney, 2009, 2011) and judgments on ensembles tend to be more accurate than those on single objects (Alvarez, 2011; Robitaille & Harris, 2011). These sorts of results have led support to claims that the processing of ensembles and single items rely on different systems (Cohen et al., 2016; Leib et al., 2012; Rhodes et al., 2018). Some mechanisms may be uniquely relevant to ensemble perception judgments. For instance, people are better at mean judgments when primed with parallel visual search (distributed attention), but better at individual item judgments when primed with serial visual search (focused attention, Chong & Treisman, 2005). Pooling mechanisms acting in the periphery lead to visual crowding and are related to ensemble perception (Parkes et al., 2001), but cannot account for single item processing (Rosenholtz, 2020).

However, many ensemble perception tasks are versions of object recognition, working memory, or attention tasks. It would seem reasonable to expect some shared mechanisms on this basis alone. For instance, judgments about both mean and individual sizes use the conceptual representation of size, and are well described on the psychological scale derived from individual item size judgments (Chong & Treisman, 2003; Teghtsoonian, 1965). Judgments about ensembles and about single objects also correlate with each other (Haberman et al., 2015; Sweeny et al., 2015). In choosing to extend the study of object recognition abilities to ensemble perception, we acknowledge that there is evidence for both common and separate mechanisms, and hope that our results may help clarify just how distinct the relevant abilities may be.

The study of individual differences in high-level visual skills is relatively young (see Gauthier, 2018 for a review) and remains limited by the number of available tasks with sufficient reliability to study individual differences (Gauthier, 2020; Hedge et al., 2017). For our purposes, ensemble judgments meet the dual requirement of being tasks that are very different from those previously used to measure O, and of producing measurements with good reliability. For instance, Haberman et al. (2015) related performance in tasks where subjects reported the average of a feature in an array of four objects or the value of that same feature for a single cued object. Average ensemble identity or emotion judgments about sets of faces were strongly correlated with similar judgments about individual faces. Chang and Gauthier (2020) designed similar tasks with familiar categories (planes, birds and cars) and found good reliability and common variance. Based on such work, we chose to explore the contribution of O to novel tasks by including judgments of the average identity of object ensembles.

Interestingly, the manner in which these ensemble judgment tasks are designed may interact with object familiarity. Tasks measuring the ability to judge the average identity of complex objects are usually conducted within a circular morph space created on the basis of a few identities. With familiar objects, the use of morphs can lead to images of instances that may appear plausible, but could not actually exist (e.g., there are no birds that are a morph of two other species, or a car that is a morph between two existing models). This may interfere with judgments for familiar objects more than those for novel objects. In addition, the small space of similar images created by morphing between (necessarily) similar instances means that very little of the entire familiar domain is sampled. Morphing images of complex natural objects is difficult, requiring the selection of images that have all corresponding features. For this reason, ensemble perception studies that use morphing, including ours, tend to use a set of morphs from a small set of images rather than creating a large number of morphs from many instances. As a result, ensemble judgment tasks repeatedly use a few highly similar images in a very small part of "bird space." This could also limit the manner in which our ensemble tasks tap into familiarity with a domain. As a result, it is possible that any familiarity effects in individual object recognition tasks is substantially reduced for ensemble judgments in tasks such as we have described.

Relation to Richler et al. (2019)

The present work is a conceptual replication and extension of a study by Richler et al. (2019). These authors used structural equation modeling (SEM) to characterize recognition abilities in 246 subjects tested with five novel object categories and three different tasks (a memory test, a matching task and a composite task (matching of object parts). A higher order latent factor, O, accounted for a large portion of the variance shared between categories and tasks. This factor accounted for approximately 89% of the variance in the lower-order factors representing each novel object category. A second experiment showed o to be largely separable from constructs like fluid intelligence, perceptual style and visual short-term memory, as well as from conscientiousness. Two results from the first experiment speak to whether O is relevant for familiar objects. First, subjects were provided with about 90 minutes of training with each of 4 of the 5 novel categories. This training was sufficient to result in holistic processing for the 4 trained categories, an effect associated with expertise (Chua & Gauthier, 2020). But surprisingly, O was equally relevant whether objects were entirely novel or familiar enough to be processed more holistically (in what is considered a more “face-like” manner; Farah & Tanaka, 1993). Second, familiar object recognition (for faces, birds, butterflies, cars, houses and planes) was measured, although only with a single task that was a memory test (rather than latent constructs with several tasks as for novel objects). The correlations between these measures and *o* ranged from .27 to .60, with faces and cars showing the smallest correlations, consistent with prior work. Altogether, these results are somewhat equivocal with regards to the relationship between and familiar object recognition.

This work is an effort to clarify the theoretical reach of O within the realm of all the possible tasks relevant to high-level vision, within the constraints of psychometric feasibility (that is, there is a small, but growing, example of object recognition tasks that are reliable enough for individual differences research). It is notable that prior efforts to find domain-general variance in visual tasks, for instance among performance between different visual illusions, failed to find evidence for a common factor, despite good reliability and even for illusions that have similar explanations (Grzeczkowski et al., 2017). Other studies in vision have also failed to find a common factor (Bosten & Mollon, 2010; Cappe et al., 2014). If O was found to be relevant across types of categories and across tasks that have been used to suggest the need for different systems or mechanisms (e.g., Brady et al., 2015; Cohen et al., 2016; Curby et al., 2004; Leib et al., 2012; Rhodes et al., 2018), it would support a more unified approach to the study of high-level vision.

In the present study, we did not train subjects with the novel objects (as in Richler et al., 2019), because (1) this manipulation did not influence the loadings on O in the previous work and (2) we wanted to maximize the difference between the novel and familiar categories. Of the three tasks used in the 2019 study, we kept two that were very different in their task requirements (the learning/memory task and the matching task) and added the ensemble perception task.

Method

Subjects

Subjects were recruited through the Vanderbilt SONA subject pool as well as through flyers posted throughout Vanderbilt campus. Sample size reflected a combination of practical considerations, including how many people could be recruited for this multi-session study in a reasonable time frame (factoring in some small degree of attrition or missing data before completing analyses), and *a priori* power analyses performed prior to the Richler et al. (2019)

study which indicated sufficient power to detect misspecifications under a range of reasonable parameter values with a sample size of 250 or greater. Two-hundred-and-ninety-four Vanderbilt University community members were recruited (100 male, 194 female; mean age = 22.6, SD = 6.5, range = 18-63). Subjects were compensated with either course credit or pay ($15/hour). In the end, as discussed in the Results section, data from 284 subjects were used.

Stimuli

The intraclass correlations of the data collected by Richler et al. (2019) revealed diminishing returns in the estimated reliability of averaged standardized scores after three categories. Therefore, the present study used three novel and three familiar object categories. For novel object categories, we chose three of the novel categories used in the 2019 studies, vertical Ziggerins, symmetrical Greebles and Sheinbugs (see Figure 2A), selected to be maximally distinct from each other (i.e., we did not include asymmetric Greebles and horizontal Ziggerins from Richler et al., 2019).

We selected three familiar categories according to four criteria:

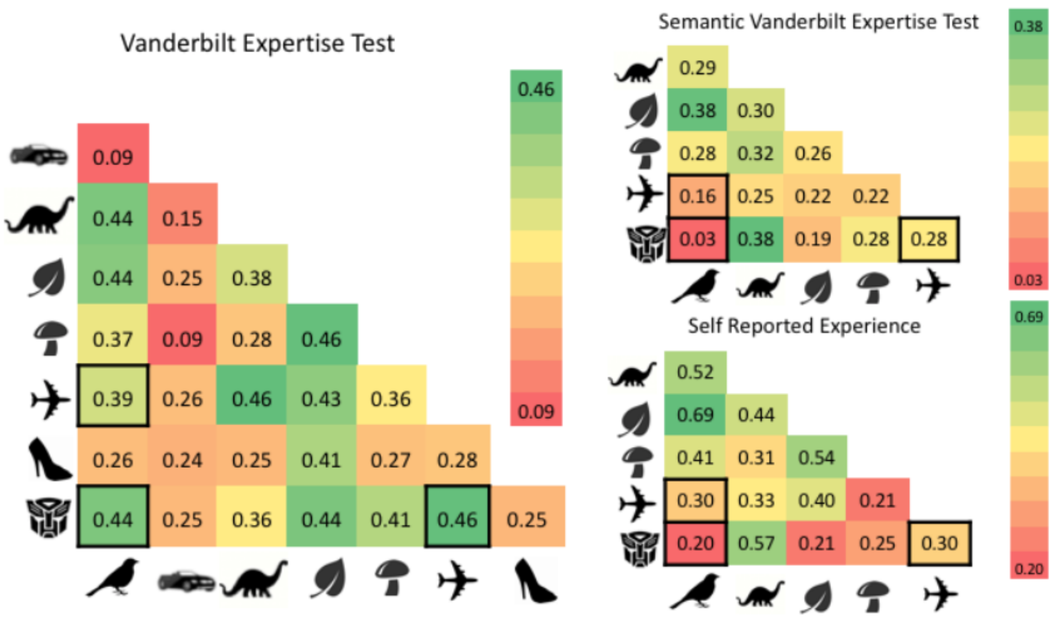
1) Avoiding categories (specifically, faces and cars) for which recognition performance appears to be “special”, that is avoiding categories for which the correlation with other categories is systematically well below the average for all pairs of familiar objects that have been considered in prior work (see Ćepulić et al., 2018; Hendel et al., 2019; McGugin et al., 2012; Van Gulick et al., 2016). Readers interested in a discussion of why car and face recognition may be special are directed to Sunday et al. (2019).

2) Choosing at least some categories with high naturally-occurring variability in experience in the population, as this has been the main explanation for the limited correlations among performance with different familiar categories. Because variability in experience is a property of most familiar object categories, we want to know whether this will impede the estimation of O_F.
3) Avoiding pairs of categories that are strongly correlated in terms of domain-specific experience. An example would be cars and planes, assuming that people interested in cars also tend to be interested in planes. In doing so, we were better able to assess individual differences in ability while pruning out the effects of experience.
4) Choosing categories that, all else being equal, represent a variety of different familiar categories (e.g., not all animals, or all large objects).

To address the first criterion, an examination of previous data relating performance on VETs for several categories (Van Gulick et al., 2016) found that recognition performance for birds, planes, leaves, dinosaurs, mushrooms and Transformers (toy robots) showed relatively high correlations with each other, as compared with categories like cars and shoes, which correlated to a lesser extent with other categories (Figure 1, left). To address the second criterion, we chose Transformers because we saw considerable variability in reported experience for such toys in previous work (Ryan & Gauthier, 2016). In addition, moving to the third criterion, Transformers, birds and planes as a group did not show correlations in self-reported experience (Figure 1, bottom right) or semantic knowledge (Figure 1, to right) higher than .3 (Van Gulick et al., 2016). As a set, these three categories satisfy the fourth criterion: with birds as physically-small, living objects, planes as physically-large, non-living and objects and Transformers as small, non-living objects with an animate shape. We acknowledge that this selection reflects compromises and was constrained by availability of prior results. All familiar object images were obtained from the Internet and include the exemplar object and background (except on the ensemble perception task, in which the background was removed). Bird images include passerine bird species common to North America, Transformer images include images of Transformer action figures (in their standing robot form rather than their vehicle form), and plane images include a range of commercial and military plane models (Figure 2A).

Figure 1

Correlations between VET Performance (left), SVET Performance (right top) and Self-Reported Rxperience (right bottom) from Van Gulick et al. (2016)



Note. N = 213. Categories are represented by silhouette figures (Vanderbilt Expertise Test (VET): cars, dinosaurs, leaves, mushrooms, planes, shoes transformers and owls; Semantic Vanderbilt Expertise Test (SVET) and Self-report: dinosaurs, leaves, mushrooms, planes, transformers and owls). Cars and shoes are not shown in right two images because they did not meet our Criterion 1 (see VET results on left). Correlations between categories chosen for the present study are outlined in black.

Tasks

Over two two-hour sessions, subjects completed 18 tests[1], with three tasks (see Table 1) for each of 6 categories, in the same order (trial order within tests was also fixed, to avoid confounding individual differences with order effect variability). The first task, which was also used in the Richler et al. (2019) study, was a learning exemplar (LE) task. This task employs a similar paradigm to several face and object memory tests (Dennett et a., 2012; Duchaine & Nakayama, 2006; Richler et al., 2017) and included a learning component over the entire course of the task (learning six exemplars which are then repeatedly tested). The second task was a same/different matching (MA) task also used in the Richler et al. (2019) study. The last task was an ensemble perception (EP) task modeled after previous work investigating individual differences in varying visual domains (Haberman et al., 2015). In Session 1, the tasks were as follows: EP-Greeble, LE-Bird, MA-Transformer, EP-Plane, LE-Sheinbug, MA-Bird, MA-Greeble, EP-Ziggerin, EP-Transformer. In Session 2, the tasks were as follows: MA-Ziggerin, LE-Transformer; EP-Sheinbug, MA-Sheinbug, LE-Plane, MA-Plane, EP-Bird, LE-Greeble and LE-Ziggering[2]. All tests were administered using Matlab and Psychtoolbox-3 software (Brainard, 1997).

[1] We refer to a task as a generic set of rules and requirements and to a test as a fixed series of trials for a given task and category.
[2] We also included three short tests of semantic knowledge for the familiar categories (Van Gulick et al., 2016) that we do not use in any way. These semantic measures were included in case we found that we could not estimate an O_F and needed an independent way to model experience. This did not prove to be necessary and the tests were not used for any other purpose because they had low reliability (α of .5 or less).

Table 1
Task Details of the Four Tasks used in the Present Study

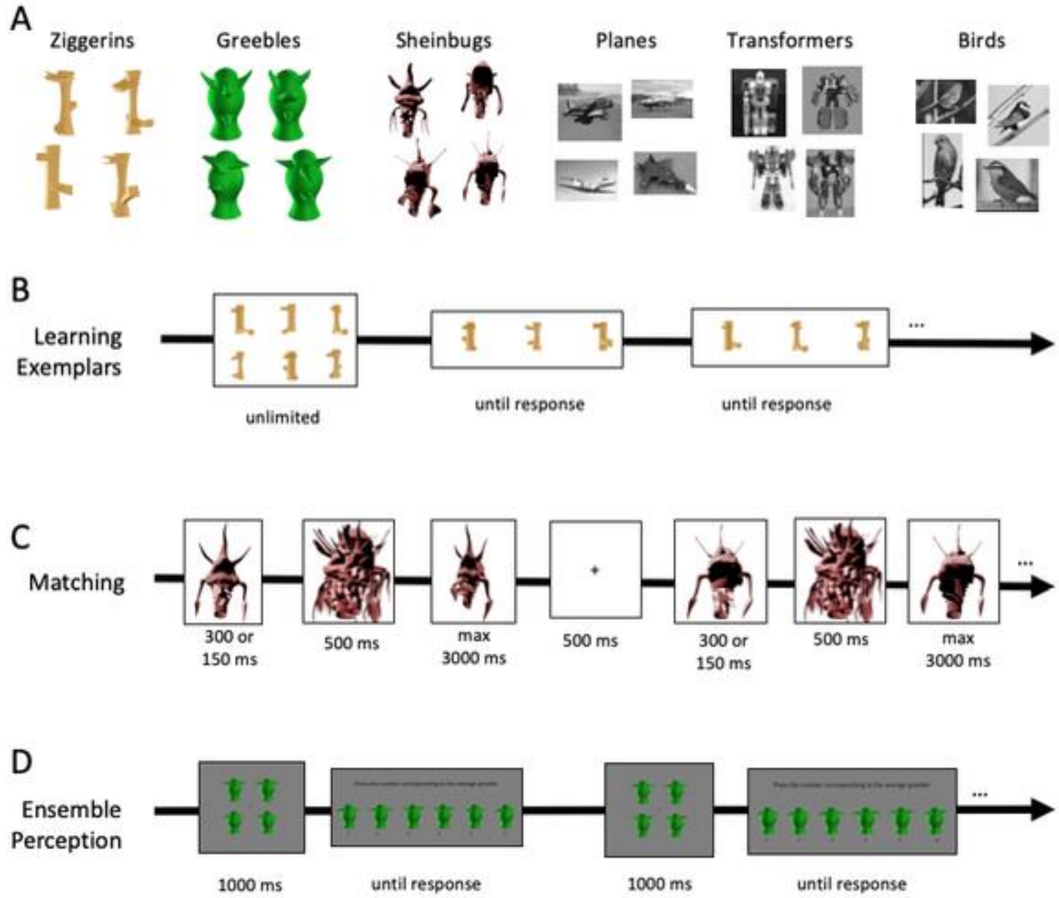
Task	Description	Trial Requirements	Response Options	Study time	Used in Richler et al, 2019 paper
LE	Study 6 exemplars that are repeatedly tested in subsequent trials	Choose studied exemplar from 2 distractors	3 AFC	unlimited	Yes
MA	Determine if pairs of exemplars are the same or different	Respond same or different to probe stimuli within 3000 ms	2 AFC	300 ms or 150 ms	Yes
EP	Estimate average identity of 4 exemplars	Choose option closest to estimated average	6 AFC	1000 ms	No

Note. LE = Learning Exemplar, MA = Matching, EP = Ensemble Perception, AFC = Alternative Forced-Choice.

Learning exemplars (LE) task

Each LE test required subjects to learn and subsequently recognize six exemplars from a given category. The three novel category LE tasks were from Richler et al. (2019) and the three familiar LE tests were originally used in Van Gulick et al. (2016). The tests were almost identical with minor differences between the familiar and novel objects. For novel objects, subjects first studied six exemplars from a given category for as long as they liked. They then completed two 24-trial blocks (48 trials total) in which they made an unspeeded choice as to which of three objects presented together were one of the six studied exemplars (Figure 2B). In the first block, the objects were shown in the same viewpoint as during study, while they were in a new viewpoint for the second block. Subjects were allowed to review the target exemplars shown together in a single display after trials 6 and 24 and instructed after trial 24 that the subsequent targets would differ in viewpoint. For familiar objects, the first 12 trials showed images in the studied viewpoint (identical images, including background), and the next 36 trials showed images that were different examples of the same objects (different viewpoint, but also different background). This task required about 8 min per category. For both tests, performance was indexed by percent accuracy over all 48 trials.

Figure 2
Stimuli and Tasks



Note. A Examples of the novel and familiar objects used in the study; B-D: Example trials for each task: learning exemplars with Ziggerins (top), matching with Sheinbugs (middle) and ensemble perception with Greebles (bottom); D: the words on the response display are “Press the number corresponding to the average Greeble.”

Matching (MA) task

This task required subjects to determine, on each trial, if a single probe object matched the identity of a single studied object, when presented sequentially. Each trial began with an object presented for 300 ms in the first block and 150 ms in the second block. This object was followed by a domain-specific scrambled mask for 500 ms, followed by a probe object and a 500 ms fixation inter-stimulus interval (Figure 2C). Subjects determined if the probe object was the same or different identity from the studied object (regardless of viewpoint), within a maximum of 3000 ms to respond. There were 360 total trials and subjects were given a break every 90 trials. There were equal numbers of same and different trials. The availability of different images for novel objects (created systematically with computer software) and for familiar objects (where different images were collected for the same identity) required some differences in how the trials were designed.

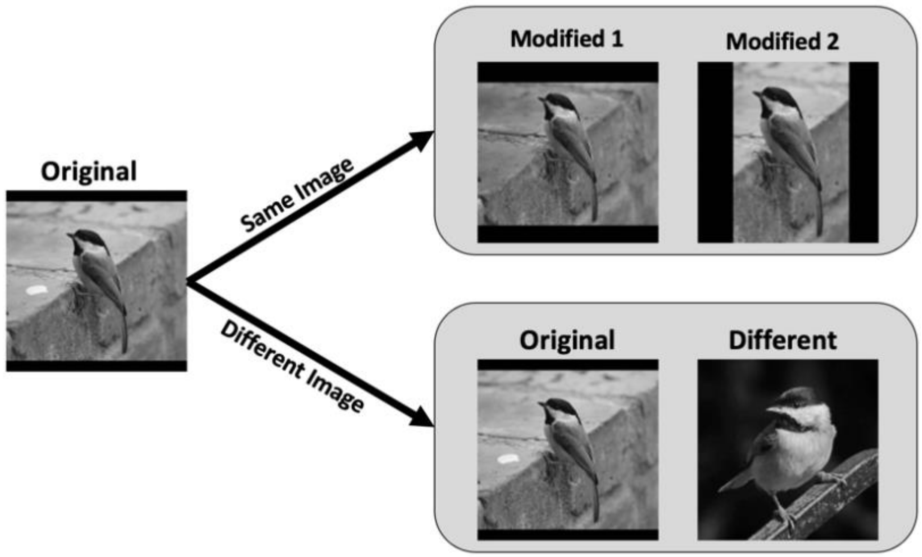
For the novel categories, the probe image size also varied randomly across trials, where half the trials presented the probe image in the same size as the study image (125x125 pixels) and half in a slightly smaller size (95x95 pixels). Fifty total objects were used, each in two different viewpoints of the objects rotated around a vertical axis (i.e. rotated out of the screen, see Figure 3, left trial). Objects repeated between the matching and learning exemplar task (70 objects for Ziggerins, 68 for Greebles, 72 for Sheinbugs).

For each familiar category, 2 views of 80 exemplars from each category were used. Some of the exemplars were also used as distractors in the LE task. Half of the trials in which the correct response was “same” showed one view of the exemplar at study and a different view at test. For the other half of the “same” trials, the study and test exemplars were the same image. On these trials, to render the background less diagnostic, the backgrounds in both images (study and test images on these “same” trials) were slightly modified using Adobe Photoshop to create two new versions that differed in cropping and or luminance (Figure 3).

The task took approximately 20 minutes to complete for each category. Performance on this task was indexed by calculating sensitivity (*d'*).

Figure 3

Examples of Stimuli used in the Familiar Domain Matching Test Same Trials



Note. All images shown here are of a Carolina chickadee (*Poecile carolinensis*), thus correct responses for both trials are "same." The original image (left) was modified using Photoshop to create two new versions with differing backgrounds (top row, "Same Image"). Bottom row shows a trial in which the original image was shown with a different image of the same species ("Different Image").

Ensemble perception (EP) task

This task required subjects to decide which of six morphed objects from a circular continuum most closely matches the average identity of four objects presented simultaneously. To create the stimuli, three images from the same category were selected to be similar in their viewpoint and were morphed together using MorphAge software (Creaceed SPRL, 2008) to produce all six possible 25%/75% and 75%/25% morphs. Each trial began with subjects studying a 2x2 array of four morphs for 1000 ms. Next, subjects were presented with an unspeeded six-alternative forced choice in which they chose which of the six options was closest to the average of the studied array (Figure 2D). The answer choices were always presented in the same order, but starting on a different morph each time. There were 70 trials and none of the three original objects that were morphed were used in the LE or MA tests. Each response was scored as the mean absolute error in degrees (as in Haberman et al., 2015). When reporting mean EP scores, a smaller mean absolute error signifies better performance. We reversed these scores so that larger values index better performance to avoid negative correlations in analyses. The task took approximately 7 minutes per category to complete.

Analyses

We conducted confirmatory factor analyses (CFAs) to test our hypotheses. Three sets of analyses examined the factorial structure of the novel tasks (to conceptually replicate the results of Richler et al., 2019), the familiar tasks, and the combined novel and familiar tasks. The specific models tested are described in the Results. CFA is a special case of SEM that can be used to test hypotheses about latent constructs that are specified as factors (for a review, see Brown, 2015). In the present context, CFA has several advantages relative to alternative statistical approaches that might be used (e.g., exploratory factor analysis or separate regression analyses designed to test specific components of hypotheses), including the ability to: specify and test the absolute and relative fit of competing models using a variety of indices; specify factor models that posit both lower-order factors that account for correlations among observed indicators and higher-order factors that account for the correlations among lower-order factors; estimate correlations among categories that are free from the attenuating effects of measurement error and category-irrelevant variance; and specify correlated error terms or task-specific method factors that can estimate the contribution of shared methods to correlations among measures (see, e.g., Brown, 2015; Hancock & Mueller, 2006; Tomarken & Waller, 2005).

We used the lavaan R library (Rosseel, 2012) to conduct the CFAs. Script files for the analyses are available on figshare (10.6084/m9.figshare.12730796). Given the non-normality of several measures and the presence of some missing data, we used the maximum likelihood estimation with robust standard errors and a scaled test statistic (MLR) estimator. To deal with missing data, MLR uses full-information maximum likelihood (FIML; e.g., Arbuckle, 1996; Enders, 2001) to estimate parameters. This approach computes the overall likelihood of a given model as the sum of the individual likelihoods of each of the cases. To deal with non-normality, standard errors are generated using a robust sandwich estimator and the chi-square test of overall fit is computed using a scaling factor. This scaled test statistic is asymptotically equivalent to the Yuan and Bentler (2000) scaled test statistic. For comparative purposes, we also conducted analyses of several models using the robust estimation procedure developed by Yuan and Zhang (2012) that is embodied in the rsem package in R (Yuan & Zhang, 2020). In addition to accommodating non-normality and missing data, this procedure down-weights outliers using Huber weights. The proportion of the data down-weighted is determined by the user. We specified proportions of .1 (the default in rsem) and .2 for the novel and familiar correlated factor models described below which were an important core of our analyses. The parameter estimates and conclusions were for all practical purposes the same as those yielded by the analyses that we conducted using the Satorra-Bentler corrections alone. Below, we report results using the Satorra-Bentler approach alone because of the similarity in results and because the S-B approach allows for a wider array of fit indices and model comparisons, has been more commonly used in applications of SEM, and has been more thoroughly studied by methodologists.

Although we report the scaled χ^2 statistic that tests whether the target model fits exactly, it is highly dependent on sample size and only affords a binary accept-reject decision rather than an indication of degree of fit on a continuum (e.g., Tomarken & Waller, 2003). For this reason, consistent with the approach used by Richler et al. (2019), we also used the following recommended SEM fit indices to assess model fit: the root mean-squared error of approximation (RMSEA; Steiger & Lind, 1980), the standardized root mean squared residual (SRMR; Bentler, 1995), and the Comparative Fit Index (CFI; Bentler, 1990). The RMSEA is an estimate of a *parsimony-corrected* fit index because it assesses the degree of discrepancy between the observed and model-implied covariances while also penalizing for model complexity. Smaller values indicate better fit. The RMSEA is typically treated as the degree to which a model fits approximately in the population, with values < .06 typically taken to indicate close fit (e.g., Hu & Bentler, 1998, 1999). The SRMR is a measure of *absolute fit* that can be interpreted as the average discrepancy between the correlations among the observed variables and the correlations predicted by the model. Lower values indicate better fit. The CFI is an index of the *incremental* or *comparative* fit of the target model relative to a baseline model of independence in which all the covariances among the observed indicators are fixed at 0. CFI values vary from 0 to 1, with values closer to 1 indicating better fit. Given that the comparison is to the independence model, the CFI often tends to indicate better fit than the other indices. We computed both the RMSEA and the CFI using the nonnormality corrections developed by Li and Bentler (2006; see also Brosseau-Liard et al. 2012) and Brosseau-Liard and Savalei (2014), respectively. Based on simulations, Hu and Bentler (1998, 1999) recommend the following criteria for adequate fit on these measures: SMSR ≤ .08, RMSEA ≤ .06, and CFI ≥ .95.

An additional feature of the analyses was model comparisons. In some cases, models were nested in one another. Model B is nested within model A if B has identical specifications to A except that at least one parameter that is freely estimated in A (e.g., a covariance between two error terms) is restricted in B (e.g., fixed at 0). In such cases, we compared models using a scaled χ^2 difference test (Satorra & Bentler, 2001) that is appropriate when the Satorra-Bentler correction for non-normality is used. If the restrictions imposed by Model B do not impair fit relative to model A, model B is deemed acceptable and will typically be favored based on the principle of parsimony. While the χ^2 difference test has the advantage of providing a focused comparison of models, we should caution that, like the omnibus χ^2 test of model fit, decisions about rejection or non-rejection are strongly affected by sample size. In addition, not all comparisons that we conducted involved nested models. Thus, it was also relevant to take into account other measures that reward fit to the data but penalize model complexity. In addition to the aforementioned RMSEA, we report values of three Information criteria: the Akaike Information Criterion (AIC; Akaike, 1973), the Bayesian Information Criterion (BIC; Raftery, 1995; Schwarz, 1978), and a sample size-adjusted version of the BIC (SABIC; Sclove,1987)) to convey the relative fit of both nested and non-nested models. These indices reward fit but also penalize model complexity, operationalized as the number of free parameters estimated by a given model. Lower values of information criteria indicate better relative fit.

We were also interested in examining and interpreting parameter estimates yielded by our CFAs (e.g., factor loadings, correlations and covariances among factors). The complete set of parameter estimates for our primary models are shown in Section III of the supplemental materials. Good fit can be considered a necessary condition for examination and evaluation of parameter estimates.

Results

All data from 12 subjects were excluded for having more than six tests on which the subject had either below chance accuracy (50% for matching, 33.33% for VET, 90 deg for ensemble perception), reaction times shorter than 2 standard deviations below the mean reaction time for each task and category, or more than 20 timed-out trials on the matching tests. Additionally, one subject chose to leave the study during the first test of the first session and so was excluded from analysis. Thus, data from 284 subjects (96% of sample; 97 males; mean age = 22.7, age range = 18-63) were included in the analyses. Some tests were not administered (47 tests from 28 subjects, 0.8% of total data), either due to computer or human error. On matching tests in which subjects timed out (responded later than 3000 ms) on more than 20 trials, the data for that matching test only was omitted (86 tests from 45 subjects, 5.1% of matching data). Additionally, one subject was accidentally run through session 2 twice, so the second session was not used. Fourteen subjects did not return for the second session. There were no significant differences between scores on the session 1 tests (MA-Greeble, MA-bird, MA-transformer, EP-Plane, EP-Transformer, EP-Greeble, EP-Ziggerin, LE-Sheinbug, LE-Bird, SVET-Plane) for the fourteen subjects who did not return for Session 2 and the 268 non-excluded subjects who did both sessions (all ps > .30 except VET-Sheinbugs, p = .06 on which the subjects who did not return for Session 2 had a higher average performance).

Descriptive statistics for each test are reported in Table 2. Mean performances on MA and LE tasks for novel categories were similar to previous work (Richler et al., 2019). Omega coefficients (McDonald, 1999) provided evidence of acceptable internal consistency. The Shapiro-Wilks global test of normality and more specific tests of whether skewness and kurtosis values deviated from their values under normality all indicated significant non-normality on most of the measures. (Table 2). This pattern is similar to that of our previous study that used only novel objects and also found evidence of non-normality for several tests (Richler et al., 2019).

Table 2

Descriptive Statistics for Each Test.

Task	N	Mean (SD)	Range	McDonald's ω	Skewness	Kurtosis	Shapiro-Wilk
EP-Z	281	58.8 (13.8)	76.2	0.85	-0.60***	3.21	0.974***
EP-G	281	56.4 (12)	65.4	0.80	-0.57***	3.15	0.976***
EP-S	268	60.0 (15)	78	0.87	-0.23	2.65	0.991
EP-B	268	56.4 (16.2)	78	0.89	-0.58***	2.8	0.967***
EP-P	281	66.0 (13.8)	72.6	0.81	-0.34*	2.7	0.988*
EP-T	281	51.0 (15)	78	0.88	-0.63***	3.1	0.968***
MA-Z	256	1.42 (0.44)	3	0.96	-0.66***	3.97*	0.976***
MA-G	268	1.22 (0.52)	2.48	0.97	0.03	2.48*	0.992
MA-S	252	0.70 (0.34)	2.05	0.93	0.06	3.29	0.996
MA-B	267	1.95 (0.40)	2.6	0.97	-0.83***	4.45***	0.964***
MA-P	256	1.88 (0.51)	3.4	0.98	-0.85***	4.56***	0.955***
MA-T	259	1.91 (0.51)	4.52	0.98	-1.65***	11.31***	0.910***
LE-Z	267	61 (15)	67	0.84	-0.07	2.39**	0.986*
LE-G	269	60 (17)	77	0.86	0.05	2.27***	0.982**
LE-S	276	51 (12)	63	0.70	0.02	2.88	0.993
LE-B	280	70 (15)	67	0.87	-0.18	2.19***	0.979***
LE-P	269	68 (12)	65	0.81	-0.32*	3.3	0.983**
LE-T	270	76 (12)	58	0.85	-0.40**	2.56	0.978***

Note. EP = Ensemble Perception, MA = Matching, LE = Learning Exemplar. Categories: Z = Ziggerins, G = Greebles, S = Sheinbugs, B = birds, P = planes, T = transformers. EP results are mean absolute error in degrees. MA results are in d-prime units. LE results are percent correct. Skewness and kurtosis values as well as the value of the test statistic for the Shapiro-Wilk test of normality are shown. In addition, asterisks denote the results of the D'Agostino test that skewness deviates from its value under normality (0) and the Anscombe-Glynn test that kurtosis deviates from its value under normality (3). *$p < .05$, ** $p < .01$, *** $p < .001$.

Pearson correlations are reported in Table 3. Due to non-normality, we also computed Spearman rank correlations. These were on average only .006 different from Pearson correlations. In addition, we computed percentage bend correlations that down-weight outlying values from univariate distributions and skipped correlations that remove outlying values from bivariate distributions (e.g., Pernet et al., 2013). In both cases the observed correlations were very close to the Pearson values with no notable deviations (see Section I supplemental materials for the Spearman, percentage bend, and skipped correlations and their comparison to the Pearson values). Exploratory analyses by sex are also provided in Section II of the supplemental materials (the only evidence in support of a sex differences was the LE test with Transformers, on which men outperformed women).

Table 3

Zero-Order Pearson Correlations for All 18 Tests

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.	18.
1. EP-B	–																	
2. MA-B	.30	–																
3. LE-B	.18	.32	–															
4. EP-P	.45	.25	.11	–														
5. MA-P	.36	.66	.27	.14	–													
6. LE-P	.29	.36	.42	.19	.46	–												
7. EP-T	.62	.37	.10	.48	.32	.22	–											
8. MA-T	.33	.58	.25	.30	.52	.32	.38	–										
9. LE-T	.28	.30	.37	.27	.34	.52	.30	.36	–									
10. EP-Z	.56	.32	.03	.47	.29	.15	.62	.36	.21	–								
11. MA-Z	.30	.50	.25	.20	.55	.38	.31	.48	.39	.33	–							
12. LE-Z	.34	.32	.30	.23	.39	.44	.26	.29	.44	.32	.47	–						
13. EP-G	.41	.27	.15	.49	.21	.14	.45	.28	.19	.44	.26	.16	–					
14. MA-G	.24	.60	.17	.22	.56	.27	.29	.52	.34	.30	.60	.32	.21	–				
15. LE-G	.38	.32	.31	.20	.38	.43	.33	.24	.54	.22	.39	.51	.20	.39	–			
16. EP-S	.71	.32	.13	.49	.38	.24	.65	.33	.31	.54	.29	.28	.39	.24	.35	–		
17. MA-S	.33	.59	.22	.18	.63	.38	.38	.54	.37	.26	.60	.34	.20	.59	.38	.33	–	
18. LE-S	.09	.23	.25	.06	.18	.37	.12	.14	.41	.06	.21	.36	.07	.21	.35	.09	.22	–

Note: EP = Ensemble Perception, MA = Matching, LE = Learning Exemplar. Categories: Z = Ziggerins, G = Greebles, S = Sheinbugs, B = birds, P = planes, T = transformers. Sample size varies because of missing values. Sample size differ across pairs of tasks, but all correlations above .11 are significant at $p < .05$, uncorrected.

To compare the correlations between familiar and novel categories, we examined the strengths of these relations within each task (Table 4). Novel categories tend to correlate to a greater extent than familiar categories (McGugin et al., 2012; Richler et al., 2011; Richler et al., 2017; Van Gulick et al., 2016), presumably because variability in experience with familiar categories attenuates correlations between these categories (i.e. varying levels of experience with different categories drives the recognition abilities for these categories to diverge). In contrast, as indicated in Table 4, familiar and novel categories in this study had very similar patterns of correlations. This could be because the 3 familiar categories were selected according to a set of criteria (see Method) that was not relevant in prior work. Using Mplus (Muthén & Muthén, 1998-2017), for each of the three tasks we constrained the average correlation among the three novel and three familiar categories to be equal. For all three tasks, this constraint failed to significantly impair the fit relative to an unrestricted model that imposed no equality constraint (MA: $\chi_1^2 = .01, p = .91$; LE: $\chi_1^2 = .37, p = .55$; EP: $\chi_1^2 = 1.78, p = .18$).

Table 4
Correlations Between All Tests Grouped by Task

	Familiar				Novel		
	Birds	Planes	Mean		Greebles	Ziggerins	Mean
MA			.59				.60
Planes	.66			Ziggerins	.60		
Trans	.58	.52		Sheinbugs	.59	.60	
LE			.44				.41
Planes	.42			Ziggerins	.51		
Trans	.37	.52		Sheinbugs	.35	.36	
EP			.52				.46
Planes	.45			Ziggerins	.44		
Trans	.62	.48		Sheinbugs	.39	.54	

Note. Ns for each correlation range from 234-276 and all correlations are significant (all *p*s < .001). The average correlation for each task is shown in shaded columns (correlations were Fisher-transformed before averaging). EP = Ensemble Perception, MA = Matching, LE = Learning Exemplar

Do latent factors account for performance across recognition tests?

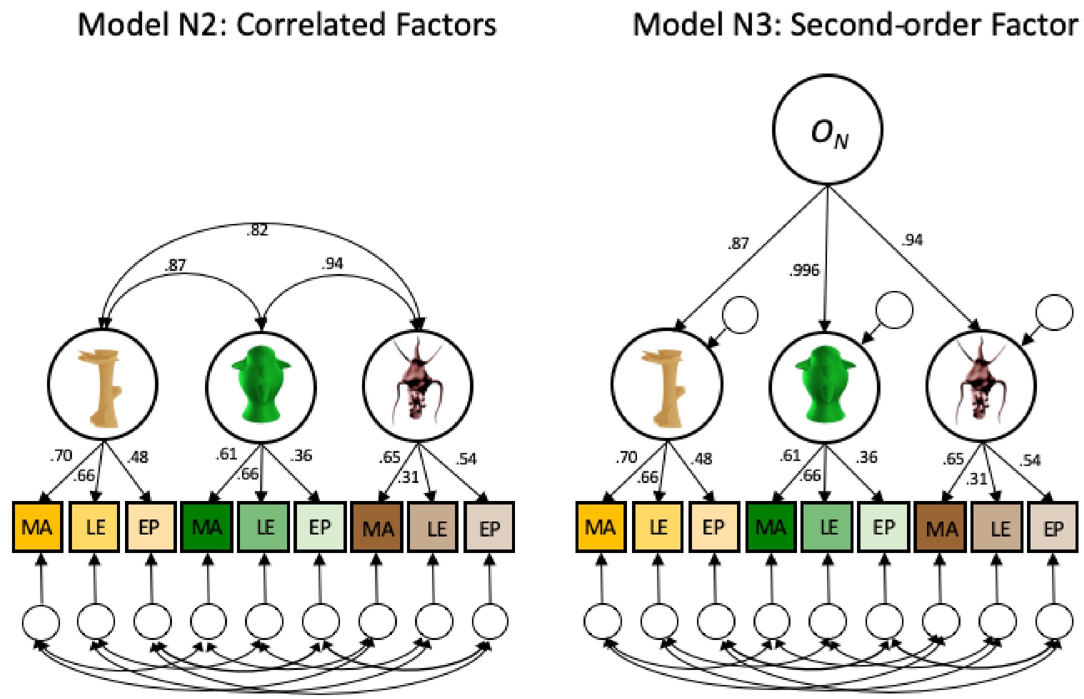
By examining the zero-order correlations, we see that the tests correlate with one-another both across-tasks and across-categories. However, these correlations can be attenuated by random measurement error and construct-irrelevant (i.e., category-irrelevant) variance. On the other hand, shared method (i.e., task) variance can inflate the within-task across-category correlations (e.g., Gauthier, 2018; Nunnally, 1970; Richler et al., 2019). Moreover, zero-order correlations only index pairwise relations between observable measures but not between lower-order or higher-order factors. To address these questions, we conducted CFAs. Two series of CFAs were fit separately for the novel and familiar categories. Then we conducted CFAs combining the data across both sets of categories.

Novel Categories

We used CFAs to test an *a priori* sequence of three models. Focusing on the novel categories, we began with Models N1 and N2. Model N2 is depicted in the left-hand panel of Figure 4. In this figure observed (i.e., directly measured) variables (e.g., MA scores) are denoted by rectangles and latent factors are denoted by circles. The small circles at the bottom are error terms that account for influences on observed scores other than the latent factor and random measurement error. The double-headed arrows connecting the lower-order factors represent covariances among the factors and the double-headed arrows connecting the errors represent covariances among the error terms that account for shared method variance within a given task. Model N1 was identical to that of N2 with the exception that no correlated within-task errors were specified. Consistent with the results of Richler et al. (2019), model N1 did not fit well (see the top block of Table 5) and, in addition, had several inadmissible estimates (factor covariances and variances that implied correlations greater than 1). This outcome is not surprising given that it is commonly necessary to specify covariances among the errors of a given task to account for task-specific method variance. N2 included correlated errors for each task. It yielded no inadmissible estimates and demonstrated excellent fit (see Table 5) that was substantially better than Model N1 according to a nested chi-square test $(\chi^2_9 = 157.93, p < .001)$ and comparison of the values of information criteria and other fit indices. The parameter estimates for model N2 are shown in the figure in the left panel of Figure 4. We omit the correlated error estimates for the sake of legibility.

Figure 4

Confirmatory Factor Analysis Models of the 3 Novel Categories (Standardized Solutions)



Note. All models specify that each of the three tasks assessing performance with a given category loads on the appropriate lower-order category factor. The directed arrows from the factors to each observed measure reflects the specification that a proportion of the variance of each observed task measure is due to the latent construct of individual differences in ability with a given category. Left: first-order correlated factor model N2. Right: second-order factor model, N3, specifying that covariances among category factors are entirely accounted for by one overarching object recognition ability (O_N). Parameter estimates shown are from the standardized solution for models N2 and N3. Parameter estimates for correlated errors are available in the supplemental materials.

Table 5

Model Fit Indices for Novel (N), Familiar (F), and Combined (C) Factor Models

Model	Description	df	χ^2	RMSEA (90% CI)	CFI	SRMR	AIC	BIC	SABIC
				Novel Categories (3 Indicators of o Categories: MA, LE, and EP)					
N1	Correlated novel categories[1]	24	176.03, *p* = <.001	.148 (.128, .169)	0.782	.082	-578.34	-468.87	-564.00
N2	N1.3ind + correlated errors across tasks	15	17.38 *p* = .30	.023 (.000, .063)	0.997	.025	-716.84	-574.53	-698.20
N3	N2.3ind + 2nd-order category factors	15	17.38 *p* = .30	.023 (.000, .063)	0.997	.025	-716.84	-574.53	-698.20
				Familiar Categories (3 Indicators of o Categories: MA, LE, and EP)					
F1.3ind	Correlated familiar categories[1]	24	205.35 p<.0000	.164 (.144,.185)	.761	.087	-504.48	-395.01	-490.14
F2.3ind	F1.3ind + correlated errors across tasks[1]	15	34.23 p=.003	.067 (.037,.097)	.975	.030	-659.78	-517.46	-641.13
F2.3ind + bounds	F2.3ind+ $-1<r_{BN,PN} < 1$ constraint[1]	16[2]	31.97 p=.01	.065 (.033,.096)	.976	.031	-659.58	-517.27	-640.94
				Familiar Categories (2 Indicators of o categories: MA and LE)					
F1.2ind	Correlated familiar categories[1]	6	93.85, *p* <.001	.204 (.169,.242)	0.753	.070	-382.00	-305.38	-371.97
F2.2ind	F1.2ind + correlated errors across tasks	4	1.01, *p* = .91	.000 (.000, 035)	1.000	.009	-452.99	-369.06	-442.00
F3.2ind	F2.2ind+ 2nd-order category factors	4	1.01, *p* = .91	.000 (.000, .035)	1.000	.009	-452.99	-369.06	-442.00
F4.2ind	F3.2ind+ Ensemble (*EP*) factor	21	30.79 p=.08	.041 (.000,.070)	.987	.029	-674.52	-554.11	-685.75
				Combined Categories					
C1	Combined N,F, and EP Factors	118	159.33, p=.01	.035 (.019,.049)	.980	.037	-2008.90	-1749.83	-1974.97
C2	C1 + equal N/F correlation with *EP*	119	160.47, p=.01	.035 (.019,.049)	.980	.037	-2009.81	-1754.38	-1976.35
C3	C2+constrain $r(O_N O_F)$=1	120	160.7, p=.01	.035 (.019,0048)	.980	.037	-2011.52	-1759.41	-1978.54

Note. χ^2 = Satorra-Bentler χ^2 test of model fit, RMSEA = robust root mean square error of approximation, CFI = robust comparative fit index, SRMR = standardized root mean square residual, AIC = Akaike Information Criterion, BIC = Bayesian Information Criterion, SABIC = sample size-adjusted BIC. Novel factor models (block 1) had 3 indicators per category factor (MA, LE, EP). Familiar models initially specified (block 2) had 3 indicators (MA, LE, EP) per category factor but consistently yielded inadmissible estimates and also had marginal overall fit according to the RMSEA.

Subsequent familiar models (block 3) had 2 indicators (MA and LE) per category factor. and model F4 added an *EP* factor that all 3 EP measures from the familiar categories load on. Combined models have 2 indicators for novel factors, 2 indicators for familiar factors, and 6 indicators for an *EP* factor.
[1] Model had at least one inadmissible parameter estimate and/or non-positive definite covariance/correlation matrix
[2] df=16 because we released a degree of freedom given that a parameter estimate was on the boundary (see, e.g., Savalei & Kolenikov, 2008)

The right-hand panel of Figure 4 depicts the parameter estimates for Model N3. In this case, we replaced the correlated factor structure in Model N2 with a second-order factor model. The correlated errors that we added in Model N2 were retained. The directed arrows from the higher-order factor (O_N) to the lower-order factors depict the influence of individual differences in a higher-order object recognition ability for novel objects on performance on the specific categories. It is important to note that the higher-order factor model is an equivalent model to the correlated factor model (e.g., Tomarken & Waller, 2003). That is, the two models impose the same restrictions on the variances and covariances among the variables. For this reason, the two models fit equivalently. Model N3 is of value, however, because it allows for an explicit computation of the proportion of the variance of the lower-order factors accounted for by the higher-order factor. As indicated by the factor loadings for the standardized solution shown in the left panel of Figure 4, such proportions are quite high: O_N accounted for 99%, 76%, and 89% of the variance in the G, Z, and S factors respectively. In this respect, the results for novel categories replicate those found by Richler et al. (2019) who used five novel categories.

Familiar Categories

The first two models that we specified for the familiar categories paralleled the left-hand panel of Figure 4, with categories of birds, planes, and transformers. Fit indices for these models are summarized in the second block of Table 5 (3 indicator Familiar Models). Similar to the CFAs performed on novel categories, we found that Model F2.3ind (add correlated error terms) improved the overall fit relative to Model F1.3ind (e.g., nested $\chi^2_9 = 166.15, p < .001$). However, Model F2.3ind was associated with an inadmissible estimate (a set of variance and covariance parameters that implied a correlation between two factors that is greater than 1). Specifically, the estimated correlation between the Bird and Plane factors was 1.03.

Two primary causes of inadmissible estimates in SEM are sampling error (i.e., the population value of the estimate is admissible) and model misspecifications (Chen et al., 2001; Kolnikov & Bollen, 2012). Given that the estimated correlations among all three category factors were high to begin with (all were greater than .90), one could argue that estimates were especially susceptible to the effects of sampling error because our sample size (*N*=284) was on the small side for a SEM study. Consistent with this argument, we could not reject the hypothesis that the correlation between the Bird and Plane factors was greater than 1.00 either by Wald or likelihood ratio difference tests (*p*s > .40).

Nevertheless, we decided to investigate systematically the possibility that inadmissibility was also caused by model misspecification. One reason was that, apart from inadmissibility, the overall fit of the correlated factor and higher-order factor models was marginal according to the RMSEA (Table 5). Secondly, a constrained correlated factors model (Model F2.3ind+bounds) imposing bounds on the correlation between Birds and Planes (i.e., -1 < *r* < 1;) still had problems (see Table 5). In this case, the covariance and correlation matrices among the three category factors were still inadmissible because they were not positive definite[3] Non-positive definite matrices also occurred when we constrained the correlation between the Bird and Plane factors to be slightly less than the boundary value (e.g., .99). That the overall *pattern* of correlations was problematic even when each individual correlation was in range suggested that factors other than an isolated case of sampling error affecting one parameter estimate alone may have played a role.

Finally, we anticipated that the ensemble task might have different patterns of correlations than the other tasks. As noted in the Introduction, we were concerned that the use of a small number of images repeated over trials in EP tests could result in insufficient coverage of a familiar domain (e.g., birds), and that morphing images may have more impact on familiar than novel objects. By altering correlations with other indicators on a domain-specific basis, this factor could lead to the violation of the model constraints that is the defining feature of a misspecified model (e.g., Tomarken & Waller, 2003).

We investigated this possibility by running three correlated-factor models in which each pair of tasks was used as the observable indicators of the B,P, and T factors. The two models that included EP as one of the indicators were both problematic: the EP-LE model had difficulty converging and appeared to suffer from empirical under-identification (e.g., Kenny, 1979)[4],

[3] Although positive definiteness is a rather technical condition (see e.g., Rousseeuw & Molenberghs, 1994, for a particularly accessible discussion of positive definite and non-positive definite 3 X 3 correlation matrices), a glance at the model-implied correlation matrices yielded by the models that we ran suggests the nature of the problem. While the markedly high correlation between the Bird and Plane factors suggests that they were nearly identical, they did not have identical correlations with Transformers: the Bird-Transformer correlation hovers around .96, while the Plane-Transformer correlation lies in the .90-.91 range.

[4] Empirical under-identification refers to those situations in which a model should be identified (i.e., estimable and testable) based on its structure and adherence to criteria for identification (e.g., Bollen, 1989), but is not identified due to the specific pattern of variances and covariance in the sample data being analyzed.

while the EP-MA model yielded an inadmissible estimate. Interestingly, although the three-indicator models fit well for the novel categories, the two-indicator models for novel categories that included EP were also problematic. The LE-EP model yielded due to some very anomalous estimates, perhaps due to empirical under-identification[5], and the MA-EP model yielded at least one inadmissible estimate. This pattern suggests that some of the issues with EP are not specific to familiar categories. As outlined in the supplemental materials, the problems involving EP appeared related to a combination of only moderate within-category correlations with other measures and higher than expected across-category correlations with other measures.

In contrast to the results for models including EP, for both novel and familiar categories, the two indicator models with only LE and MA and within-task correlated errors fit quite well (novel: $\chi^2_4 = 3.96, p = .41, \text{RMSEA=0.00,CFI=1.00,SRMR=0.017}$; familiar: $\chi^2_4 = 1.01, p = .91, \text{RMSEA=0.00,CFI=1.00,SRMR=0.014}$)[6]. The results of the two-indicator (LE and MA) models for familiar objects are found in Table 5 (models F1.2ind-F3.2ind). As was the case for the novel CFA models, the inclusion of within-task correlated errors significantly improved overall fit according to a nested test $\left(\chi^2_2 = 140.80, p < .001\right)$ and comparisons of fit

[5] When the LE-EP model was run using lavaan, convergence was achieved after a high number of iterations and there were no warnings. However, some estimates were clearly anomalous (e.g., unstandardized factor loading of s_ep on the Sheinbugs factor was over 9 times greater than the loadings of ep on the other two novel factors, the estimated variance of the Sheinbugs factor was effectively 0). Anomalies were also observed when we re-ran the model fixing factor variances at 1 to identify models rather than using fixed factor loadings. When we re-ran the model in MPLUS, we received an error message that the robust chi-square and standard errors could not be computed. Although the precise reason for the discrepant output across the two packages is unclear, the important point is that both indicated problems with the LE-EP model.

[6] To identify the two-indicator models for familiar categories, we imposed equality constraints on the correlated errors. The covariances among the errors of the three LE tasks were constrained to be equal, as were the covariances among the errors of the three MA tasks. As a result, the addition of correlated errors only reduced the degrees of freedom for models F2 and F3 relative to model F1 by 2 (6 vs. 4).

indices. The correlations among the category factors for this model were quite high (*r*s =.85, .81, and .85; mean r = .84). These values are just slightly lower than correlations among the novel factors reported above (mean r = .88). Consistent with this result, the higher-order factor model (Model F3.2ind in Table 5) indicated that a higher-order object recognition factor, (O_F) for familiar items accounted for a substantial amount of variance in the lower-order category factors (proportions of variance for Birds, Planes, and Transformers = .80, .89, and .81, respectively; see right panel of Figure 4 for a structurally identical model). Specific parameter estimates are shown in the supplemental materials.

Although the two-indicator models of O_F that eliminated the ensemble task fit well, the question remains whether there are acceptable alternative models of the familiar data that incorporate all three tasks in some fashion. To address this question, we assessed whether a model (denoted F4.2ind in Table 5) including both the ensemble task and the other two tasks could be fit if we treated ensemble tasks as indicators of a separate *EP* factor that was freely correlated with the higher-order O_F factor. This model fit well (Table 5) and all estimates were admissible. The correlation between the higher-order O_F factor (on which the Birds, Planes, and Transformers Category factors loaded) and the *EP* factor equaled .66 ($p < .001$). Comparisons of information criteria (AIC, BI, SABIC) and other SEM fit indices (e.g., RMSEA, CFI) also indicated that this model fit better than: (1) The 3-indicator models of O_F that yielded inadmissible estimates (compare the values for model F4.2ind to those of the 3-indicator familiar models shown in the second block of Table 5); and (2) Several other plausible 3-indicator models such as variants of the correlated trait-correlated methods minus 1 (CT-C(M-1) model (e.g., Eid et al., 2003)[7], the bifactor model (e.g., Reise, 2012), and a model specifying one factor on which all 9 items loaded and correlated within-task errors (cf., Wilhelm et al., 2010).[8] See the supplemental materials for more details about model comparisons.

Combined Novel and Familiar Categories

We tested a factor model (denoted C1) that combined the higher-order models for novel and familiar categories. As noted, although the familiar models with EP tests as indicators of the familiar categories did not fit acceptably, a higher-order O_F model with *EP* as a separate factor fit well. In addition, although models including MA, LE and EP as indicators of novel factors fit well, two-indicator models that included EP indicators had similar problems to the two- and three-indicator familiar-category models that included EP as an indicator of the category factors. To promote structural symmetry across the novel and familiar categories, we initially specified a combined model (C1) that included MA and LE as the two indicators of each of the six lower-order factors (see Figure 5). Model C1 also specified: (1) Higher-order O_N and O_F factors; (2) an EP factor (*ep*) on which all 6 EP measures loaded; (3) Freely estimated correlations among O_N, O_F, and *EP*; and, (4) Correlated errors among the MA and LE indicators, with three separate equality constraints on the within-novel, within-familiar, and across-higher order category correlations of each of the two measures; and, (5) Equality-constrained correlated errors among the novel *EP* indicators and among the familiar *EP* indicators.[9]

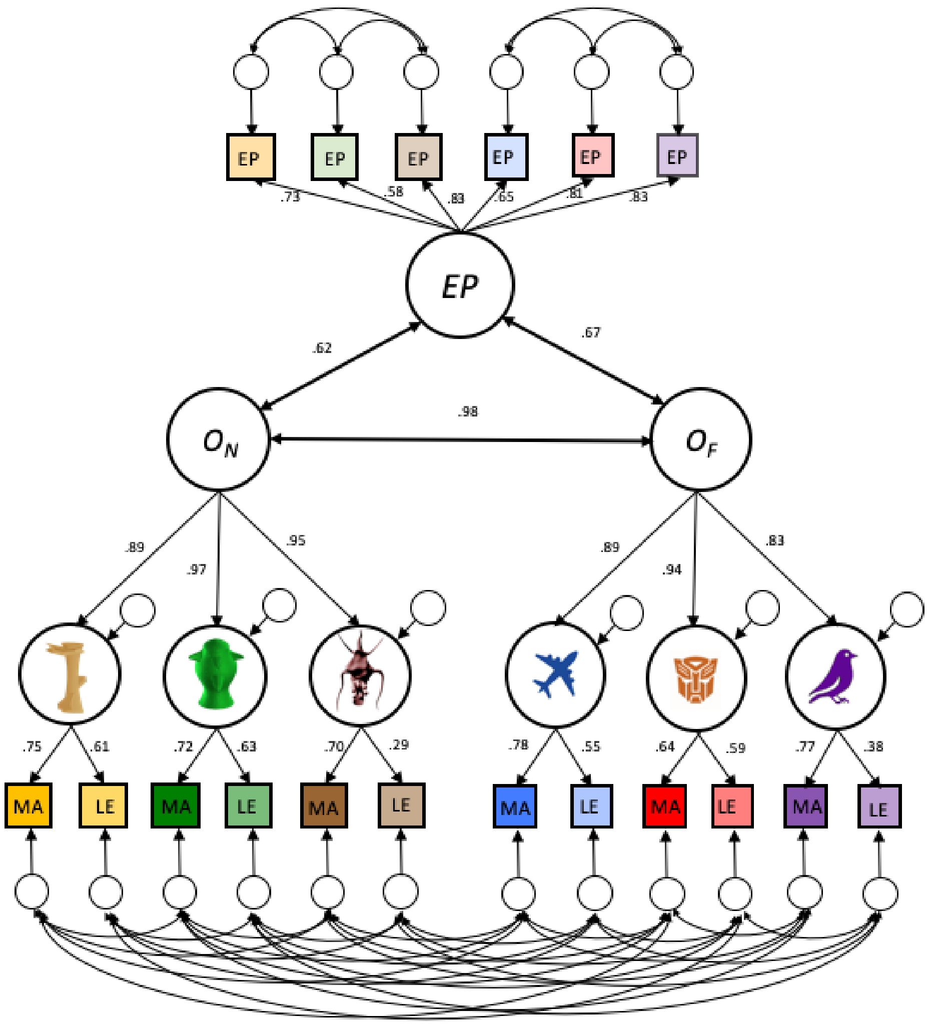
[7] As discussed in the supplement, in addition to the interpretive issues that such models pose, two of the three CT-C(M-1) models that we ran also had inadmissible estimates.
[8] We should note that selection of non-nested structural equation models based on the relative magnitude of the observed values of information criteria can be misleading due to sampling error (Preacher & Merkle, 2012). While Merkle, You, and Preacher (2016) have developed a procedure and R package (Merkle & You, 2014) for comparing non-nested SEM models based on the work of Vuoung (1989), we could not use it in the present context due to the need for non-normality corrections.
[9] Equality constraints on correlated errors were imposed for several reasons: (1) They mirrored the procedure used for the two-indicator familiar models, for which such constraints were necessary to insure an identified (i.e., testable

As indicated by Table 5 and the estimates shown in Figure 5, this model fit very well. A notable feature was the extremely high correlation between the higher order novel (O_N) and familiar (O_F) factors (*r*=.98, *p* < .001) (see Figure 9 below). Correlations with *ep* were .67 (O_F) and .62 (O_N), respectively. To test whether O_F and O_N were equally correlated with *ep* , we constrained the two correlations to be equal.[10] This model (C2) fit well in an absolute sense (see fit statistics in Table 6) and as well as Model C1 (e.g., highly comparable information criteria, nested χ_1^2 = 1.13, p =.29). By specifying only one O factor on which all 6 lower-order category factors loaded, Model C3 further constrained the correlation between O_N and O_F equal to 1. This model also fit well in absolute sense (see Table 5) and relative to model C2 based on a nested chi-square test $\left(\chi_1^2 = .25, p = .61\right)$ and comparison of fit indices and information criteria. All told, these results indicate that the latent constructs tapped by O_N and O_F are functionally identical.

model); (2) Nested chi-square tests, information indices, and other measures of fit indicated that such constraints did not impair overall model fit while providing a more parsimonious account of the data; and, (3) There was no theoretical reason to expect the magnitudes of within(across)-category correlations to differ. Combined models without equality constraints led to essentially identical results and conclusions.

[10] We parameterized the combined models in such a way that the variances of the O_N, O_F, and *EP* factors were fixed at 1 with no fixed unit-factor loading constraints (e.g., Bollen, 1989). Thus, tests performed on factor covariances were equivalent to tests on factor correlations.

Figure 5
Model C1: Correlated O_N, O_F, and *EP* Factors (Standardized Solution)



Note. Categories are represented by silhouette figures. EP = Ensemble Perception tests, MA = Matching tests, LE = Learning Exemplar tests. Note that EP indicators are colored and ordered by category, in the same manner as the MA and LE indicators.

General Discussion

This study offers a conceptual replication of the work that first proposed an *o* factor (Richler et al., 2019). With novel objects, while using more varied tasks as indicators including an ensemble perception judgment, we found that on average, 88% of the variance in lower-order factors was accounted for by a higher-order factor O_N (compared to 89% in Richler et al., 2019). O_N also accounted for 88% of the variance in lower-order indicators with only two tasks (without ensemble perception). Importantly, we extended prior results by showing that O_F also accounted for a majority (on average 79%) of the variance in lower-order factors, using familiar object categories. Strikingly, O_N and O_F were almost perfectly correlated, representing a common factor O. This was despite performance with familiar objects being tested with images that varied much more in viewpoint and illumination than those of novel objects, which also included distracting background information, and with one of the tests showing substantial sex differences.

This work illustrates the benefits of a latent variable approach over other methods to investigate the generality of a concept like *o.* Other approaches that estimate domain-general abilities by aggregating variability across different categories for a single task have suggested a shared ability. With eight familiar categories, the average pairwise correlation different memory tests was .32, but the average correlation for each of these categories with an aggregate for the other seven categories was .49 (Richler et al., 2017). Aggregating certainly improves measurement (Rushton et al., 1983) but the latent variables approach is superior because it allows the estimation of relations between variables without attenuation by random measurement error and construct-irrelevant variance. In the present case, the latent variables for O_N and O_F build on indicators that vary both in terms of the categories and in terms of task requirements. The finding that O_N and O_F are essentially the same functionally is important, but the fact that this component is not limited to processes specific to a memory or a perception task is also a testament to the breadth of O. It suggests that O may contribute to performance across a variety of other tasks (including here, ensemble perception).

Our results add to a small but growing literature using SEM to characterize high-level visual abilities. Each of these studies has limitations, contributing to the understanding of specific aspects of visual abilities, particularly those aspects measured with multiple indicators. For instance, early work focused on face cognition and characterized different facets (face perception, face memory and speed, Wilhelm et al., 2010; see also Hildebrandt et al., 2011). This work provided some of the first insights into the structure of face-processing abilities, by the inclusion of several tasks (or indicators) for each of these three factors, which were found to be distinct from general cognitive abilities and object cognition (Wilhelm et al., 2010). Because it used only one category of objects in all tasks measuring object cognition (in this case, houses), this work did not test or establish generalization across non-face object categories. Likewise, because it included only one memory task and four perception tasks for houses, this study could not speak to perception vs. memory factors for non-face objects. More recent work (Cepulic et al., 2017) supported an object recognition factor based on indicators for several object categories, distinguishing it from a face and a vehicle factor. But because the study only included memory tasks, it could not speak to whether the distinction between perception and memory observed for faces in early work also exists for objects.

All studies are limited in the number of tasks they can include. As a result, each one can only speak to distinctions that are clearly operationalized and can only generalize over dimensions that are well sampled. Although we used both perception and memory tasks, we did not include several instances of each and so we also cannot test the strength of that distinction for non-face objects. An interesting possibility is that this distinction could differ between novel and familiar objects, and it could be even stronger for categories like faces and cars, as part of the specialization of these categories noted by many (Ćepulić et al., 2017; Sunday et al., 2019). Future work could investigate this hypothesis.

While our work does not speak to object recognition abilities that are distinct for perception and memory, it provides a clear answer about those abilities that are common across perceptual and memory tasks. Such task-general abilities are the same for novel and familiar objects. We selected our familiar categories to be representative of various types of familiar objects (a living category where objects are relatively small, a non-living category where objects are relatively large, a non-living category where objects are meant to look animate). In addition, when studies focusing on a single task (the equivalent of our LE task) used these categories, they were representative of performance for other categories including butterflies, dinosaurs, fish, flowers, leaves, motorcycles, mushrooms, owls, shoes and wading birds (Ćepulić et al., 2018; McGugin et al., 2012; Richler et al., 2019; Van Gulick et al., 2015). In recent work, O predicted 12% of the variance in performance on a task measuring the ability to detect tumors in chest radiographs, after controlling for g (which itself accounted for 8% of the variance) and for experience in this domain (Sunday et al., 2018b). We are therefore confident in suggesting that O accounts for performance across a wide range of familiar object categories and that it can account for performance on important tasks, in both novices or experts.

Ensemble Perception

Because of suggestions that ensemble perception relies on mechanisms different from those of object recognition (Cohen et al., 2016; Leib et al., 2012; Rhodes et al., 2018), tests that tap into this ability provided a strong test of the generalization of O across tasks. At least with novel objects, we were able to estimate O using ensemble perception together with matching and learning tasks, in a model with excellent fit that replicated prior work. However, ensemble perception tests proved challenging for several of our models and in the end, we had more success including a separate *EP* factor reflecting common variance across all of our EP tests with both novel and familiar objects. With this approach, we found that O shares about 42% of the variance with *EP.* This of course leaves substantial residual variance across categories, which may represent the separate mechanism suggested by some authors.

The judgments of average identity task we used here were modeled after the work of Haberman et al. (2015). These authors measured ensemble judgments with faces in identity and facial expressions conditions, and with ensembles of simple stimuli where judgments applied to more basic dimensions (such as color or orientation). Based on the pattern of correlations across tasks, they suggested that ensemble perception abilities are separated into one mechanism for high-level properties and another for low-level properties. When ensemble judgments are measured with orientated bars or colored circles, individual differences are expected to tap almost entirely into an ensemble perception mechanism (as we would expect little variability for such basic domains). With more complex objects, especially familiar domains where experience can contribute to performance, it makes sense to ask how much of a latent construct for “bird recognition” contributes to ensemble judgments with birds.

Our results suggest that O contributes to EP judgments with non-face objects, whether they are one of several indicators of abilities to recognize novel objects, or direct indicators of an *EP* factor for novel and familiar objects. But at least when tests are built with a very restricted morph space, ensemble judgments may not always strongly recruit domain-specific abilities. In addition to the small number of objects used to create the morphs in these tests, the morphs themselves may be too artificial to tap into experience with real objects. This leads to a few recommendations: if the goal is to estimate O, for example using an aggregate of several visual tasks and categories, then ensemble judgments as used here can be helpful. However, to estimate domain-specific abilities, it may be helpful to explore other kinds of ensemble tasks that do not require morphs and allow the use of large number of different stimuli (Cha et al., 2019).

Limitations and future developments

Our sample size (N=284) is large relative to most perceptual studies but small relative to most research using CFA. However, our models fit very well (low RMSEA) and the conceptual replication of Richler et al. (2019) was very convincing. While we found that O generalized to images of familiar objects in ecological settings, all these images were grayscale such that we did not learn about the variability that may come from people's ability to use color for object recognition. The range of domain-specific visual experiences people can have is much broader than the range of stimuli and tasks we used here, and future work should continue to sample this space.

It is impossible to say from our results whether O is a strictly visual ability. In recent work, we found that the ability to recognize novel objects by touch was moderately related to an aggregate of two tasks with different novel objects, chosen to estimate O (Chow et al., 2020). Future work should explore whether O can account for latent abilities in haptic object recognition, which appears possible given the extent to which haptic object recognition engages shape-selective areas in the visual system (Snow et al., 2014). This will however require the creation of several tests with adequate psychometric properties, a requirement that has limited the study of individual differences in visual object recognition (Gauthier, 2018). The technical challenges of reliably measuring individual differences with touch are even greater than with vision.

In addition, further work is necessary to assess the degree to which specific categories demonstrate factorial invariance. For example, are the factor loadings of a given task (e.g., MA) on an object category equivalent (e.g., birds vs. planes)? Exploratory analyses that we conducted based on a reviewer's comment indicated that such factorial invariance did not hold. It is important to note, however that we did not intend to address such factorial invariance in the present context because we did not explicitly match a given task (e.g., MA) on psychometric properties (e.g., task difficulty) across categories. In addition, we believe that in the long run the most productive approach may be one that incorporates item response theory (IRT) (e.g., Ayala, 2009; Embretson & Reise, 2000) for the selection of task items and modeling of data at the item level. This is a focus of our ongoing work.

Our work suggests that O_N and O_F are the same ability, but this may not generalize to populations of subjects different from the one we sampled from (adults from a university campus community). At least for "special" categories, there is documented evidence of important differences in the structure of individual differences on the basis of factors like age (Sunday et al., 2018a) or hometown population (Sunday et al., 2019). It is possible that factors like culture and socioeconomic status have an impact on the measurement of O just as they have for *g* (Ford, 2004). If so, the use of novel objects could offer less biased measurements, if differences in experience with familiar categories are a problem. Looking ahead, it is possible that visual ability testing could help place minority on equal ground with the majority culture in predicting aptitude for a variety of occupations, similar to the addition of creativity testing to standard tests (Sternberg et al., 2006). However, this should not be merely assumed, given evidence that even psychometrically unbiased tests for intelligence like the Raven's matrices are nonetheless not culture blind (Owen, 1992). These are important challenges for the growing field of individual differences in visual cognition.

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